Whistle bn-BD: 2-bit vs 4-bit

Cactus CPU inference (cactus-needle 3.2.1, libneedle) · eleven test sets · lower is better for every number

ARCHIVE SIZE	MEAN TEST WER	FLEURS VALIDATION WER	CPU PER CLIP, ZEN 5, 1 THR
18.07 MB	**30.70%**	**19.41%**	**2,821 ms**
4-bit 30.5 MB, 41% smaller	4-bit 30.05%, +0.65 pt	4-bit 18.86%, +0.55 pt	4-bit 499 ms, 5.7× slower

Large figures: the 2-bit archive; the 4-bit archive below each.

Summary

The 2-bit archive is 18.07 MB against 30.5 MB for the published 4-bit one (41% smaller). Accuracy drops a little: the mean word error rate over the eleven test sets goes from 30.05% to 30.70% (+0.65 points), and the 2-bit model is worse on ten sets, by at most 1.11 points (Common Voice 17), and better on Ben-10 dialects. On x86 CPUs it is much slower: 5.7× on one Zen 5 thread, 5.2× on four, and 4.2× / 2.8× on Cascade Lake. All of the slowdown is in the audio encoder and prefill (time to first token, 8.1–13.1× longer); decoding costs about the same per token for both archives. ARM CPUs, where the engine's 2-bit kernels may behave differently, were not tested. The 2-bit archive suits a tight storage or download budget; the 4-bit archive remains the faster and slightly more accurate choice on x86.

Table 1 · Word and character error rate per test set (%)

		WER			CER		
Test set	clips	4-bit	2-bit	Δ	4-bit	2-bit	Δ
FLEURS	920	19.17	19.53	+0.36	6.27	6.76	+0.49
SUBAK.KO	1,000	18.79	19.55	+0.76	7.89	8.08	+0.19
BanSpeech	1,000	31.54	32.09	+0.55	14.02	14.33	+0.31
Common Voice 17	1,000	17.10	18.21	+1.11	5.45	5.89	+0.44
Ben-10 dialects	1,000	79.87	79.59	−0.28	53.81	53.03	−0.78
Vaani	1,000	27.43	28.38	+0.95	9.54	10.08	+0.54
SPRING-INX R1	1,000	43.38	44.06	+0.68	26.05	26.81	+0.76
SPRING-INX R2	1,000	42.19	43.05	+0.86	24.41	24.95	+0.54
Kathbath known	1,000	15.61	16.28	+0.67	3.68	3.91	+0.23
Kathbath unknown	1,000	15.82	16.37	+0.55	3.59	3.89	+0.30
IndicVoices	1,000	19.62	20.63	+1.01	8.16	9.08	+0.92
Mean of the eleven sets		**30.05**	**30.70**	**+0.65**	**14.81**	**15.16**	**+0.35**

Δ = 2-bit minus 4-bit, in percentage points; positive means the 2-bit archive makes more errors. The 4-bit column is the benchmark's published Whistle bn-BD CPU row. FLEURS has 920 test recordings; the other sets are the benchmark's seeded 1,000-recording samples of each official split (Ben-10: validation split). The mean is unweighted over the sets. Checkpoint selection used FLEURS validation (402 clips): 4-bit 18.86 / 5.97, 2-bit 19.41 / 6.31 WER / CER.

Keep in mind

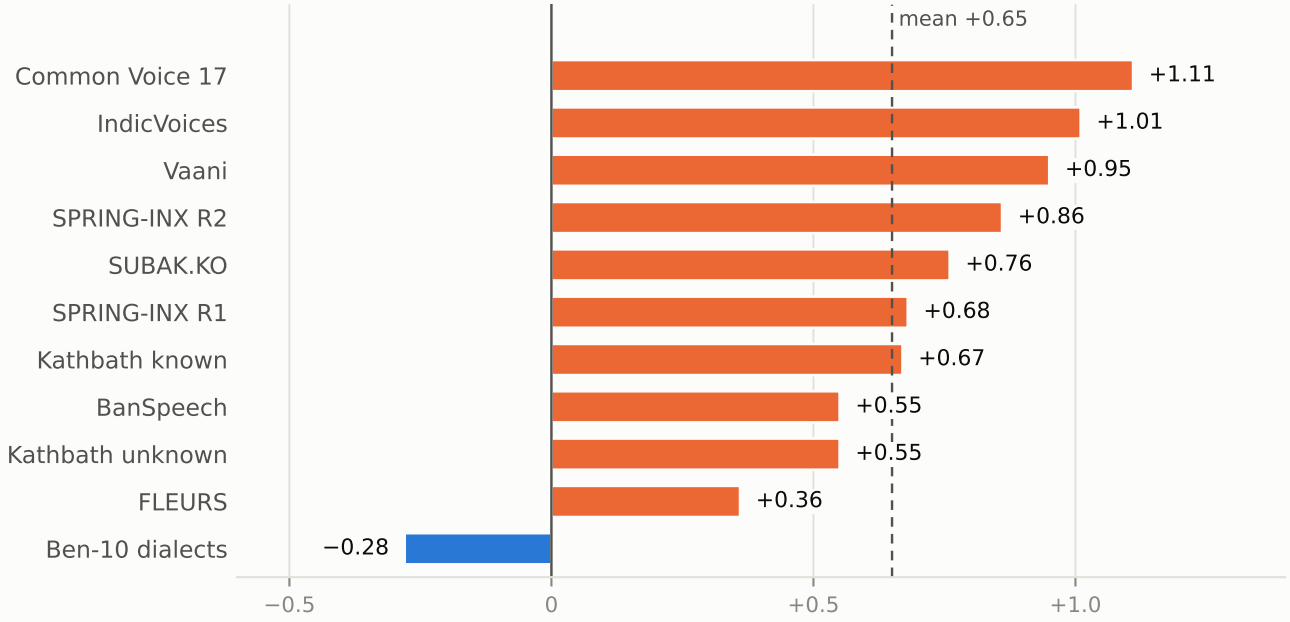
- Speed was measured on x86 server CPUs only. Phones and other ARM devices, the engine's main target, were not tested, and the 2-bit/4-bit speed ratio there may differ.
- The per-set differences are small next to the spread between sets; no confidence intervals are given for the difference.

Where the 2-bit archive costs

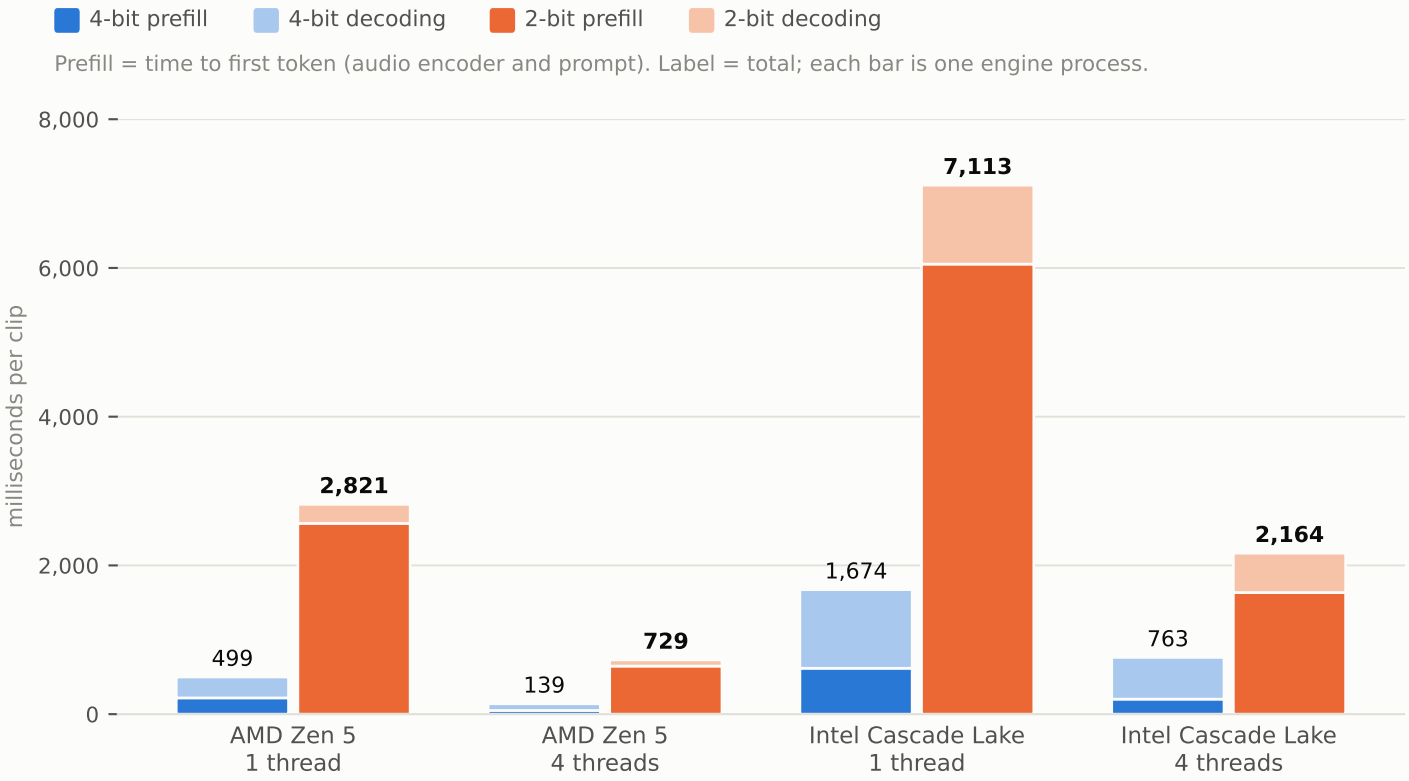
Accuracy change per test set, and CPU time per clip split into prefill and decoding

Change in word error rate, 2-bit minus 4-bit (points)

Orange bars: 2-bit worse; blue bar: 2-bit better. Dashed line: mean change.



CPU time per clip (ms), 101 FLEURS validation clips



Speed table and method

Same machine, same clips and same engine for both archives on each CPU

Table 2 · CPU speed, one engine process

CPU	threads	archive	ms/clip	prefill ms	decode ms	ms/token	RTF
AMD Zen 5	1	4-bit	499	218	281	5.19	0.037
		2-bit	2,821	2,565	257	4.73	0.210
	4	4-bit	139	50	90	1.65	0.010
		2-bit	729	646	83	1.53	0.054
Intel Cascade Lake	1	4-bit	1,674	617	1,057	19.52	0.124
		2-bit	7,113	6,052	1,061	19.60	0.528
	4	4-bit	763	202	561	10.37	0.057
		2-bit	2,164	1,636	528	9.76	0.161

Means over 101 FLEURS bn_in validation clips (about 13.5 s of audio each), both archives run one after the other on the same machine. Prefill = the engine's reported time to first token (audio encoder and prompt); decode = the rest of the clip's time; ms/token = decode time over the estimated output tokens (about 54 per clip for both). RTF = processing time over audio duration. CPUs: AMD Zen 5 server CPU (AVX-512); Intel Cascade Lake server CPU (AVX-512, no VBMI). One pass per row; in a two-pass check on Zen 5 (the probe below) passes differed by under 1%.

How it was measured

- Both archives are Cactus Compute .cact files of the same Whistle bn-BD model, run by Cactus's own CPU inference engine (cactus-needle 3.2.1, libneedle), not the PyTorch or GPU paths used elsewhere in the benchmark.
- 4-bit: the published 30.5 MB archive, every quantised tensor at 4 bits. 2-bit: 18,073,118 bytes; the embedding and mhc_phi tensors stay at 4 bits and the other 145 quantised tensors are 2-bit, the same layout as Cactus's own English 2-bit Whistle archive. The 2-bit model was fine-tuned for 2-bit weights and selected on FLEURS validation.
- Same engine settings for both: the engine's default decoding (its beam search), language bn, no external language model, no decoding options changed.
- Same audio and scorer as the benchmark: 16 kHz mono, cut at pauses into pieces of at most 25 s by the benchmark's audio path, piece texts joined with a space; punctuation removed and Bengali text Unicode-normalised before counting errors.

Mixed-layout probe (not fine-tuned)

To locate the slowdown, the 4-bit model was also exported with only one half at 2 bits and timed on one Zen 5 thread (41 FLEURS validation clips, mean of two passes). These archives were not fine-tuned, so only their speed and size are meaningful.

layout	size MB	prefill ms	ms / clip
all 4-bit (published)	30.5	202	486
all 2-bit (direct export, not fine-tuned)	18.1	2,371	2,694
encoder 4-bit, decoder 2-bit	21.5	658	934
encoder 2-bit, decoder 4-bit	27.1	1,919	2,244

A 2-bit encoder is what makes prefill slow (1,919 ms against 658 ms with a 4-bit encoder). Keeping the encoder at 4 bits and the decoder at 2 bits gives a 21.5 MB archive at 934 ms per clip; its accuracy after fine-tuning was not measured.